Matthew S. Scott

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Education

PhD in Mathematics, *University of British Columbia* 2023-2027 (expected)
Supervisors: Yaniv Plan, Özgür Yılmaz

MSc in Mathematics, *University of British Columbia* 2021-2023
Thesis: *Generative Compressed Sensing with Fourier Measurements*. Grade average 92%.

BSc, Joint Honours Mathematics and Physics, *McGill University* 2017-2021
First Class Honours; minor in Computer Science; GPA 3.89/4.00.

DEC, Honours Pure and Applied Science, *Marianopolis College*, Montreal 2015-2017

Research

Compressed sensing with generative neural networks: high-dimensional probability, inverse problems, sample complexity and sampling schemes, robustness to noise, complexity of neural networks.

- Y. Plan, **M. S. Scott**, O. Yilmaz, *Exact signal recovery from generative priors with sigmoid output activation*, submitted to Sampling Theory, Signal Processing, and Data Analysis, Oct 2026.
- Y. Plan, **M. S. Scott**, O. Yilmaz, *Partially deterministic sampling for compressed sensing with denoising guarantees*, submitted to Information and Inference: A Journal of the IMA, Mar 2026.
- Y. Plan, **M. S. Scott**, X. Sheng, O. Yilmaz, *Denoising guarantees for optimized sampling schemes in compressed sensing*, SIAM Journal on Mathematics of Data Science, Jun 2026.
- A. Berk, S. Brugiapaglia, Y. Plan, **M. S. Scott**, X. Sheng, O. Yilmaz, *Model-adapted Fourier sampling for generative compressed sensing*, Oral at NeurIPS 2023 Workshop on Deep Learning and Inverse Problems, Dec 2023.
- A. Berk, S. Brugiapaglia, B. Joshi, Y. Plan, **M. S. Scott**, O. Yilmaz, *A coherence parameter characterizing generative compressed sensing with Fourier measurements*, IEEE Journal on Selected Areas in Information Theory, Sep 2022.

Convex optimization: duality, convex separation, facial reduction, average-case analysis.

- **M. S. Scott** (sole author), *Convexity from order: a theory of faces and generalized separation*, major revisions at Journal of Convex Analysis, Nov 2026.
- **M. S. Scott** (sole author), *Bilateral facial reduction: qualification-free subdifferential calculus and exact duality*, major revisions at Set-Valued and Variational Analysis, Aug 2026.
- M. P. Friedlander, S. Kubal, Y. Plan, **M. S. Scott**, *Average-case thresholds for exact regularization of linear programs*, minor revisions, second round at SIAM Journal on Optimization, Aug 2026.

Work experience

Teaching Assistant, Research Assistant and Small Class Instructor, *University of British Columbia* 2021-present

Freelance Tutor, *Superprof*, Montreal 2020-2021

- Rated 4.87/5 over 15 reviews. Earlier: math tutor at Paramount Study School (2017) and Collège Notre-Dame (2015).

Research Assistant in Computer Science, *McGill University (Prakash Panangaden)* Summer 2020

- Computational improvements to Yaida's FDR optimizer, and application to actor-critic RL.
- Funded by an NSERC USRA.

Machine Learning Intern, *Decathlon Canada*, Montreal Summer 2018

- Built a recurrent-network sales forecaster that beat 4 of 5 in-house models, the only one predicting long term.
- Built a Python (Flask) API ranking sport popularity from Google search data.

Web Application Developer, *GoRush*, Montreal Summer 2017

- Sole developer of an on-demand lawn-mowing app (JavaScript, Meteor, MongoDB).

Software projects

- **Latex Exporter**, 4,700+ downloads, 44 stars (TypeScript). An Obsidian plugin that infers a paper's LaTeX structure from the note graph: wikilinks become citations or references, embedded notes become theorems with proofs.
- **Cracker Craft**, 10k+ Google Play Store downloads (2016; C#, Unity). Design your own firework!

- **Experiments on Slurm** (Python, Bash). Experiments as a data frame, one row per run: input columns set the variables, output columns are filled by running the rows in parallel on a Slurm cluster, ready to plot.
- **Research numerics** (PyTorch, pandas, seaborn). Numerical experiments for signal recovery with sparse and generative priors.
- **Line-following robot** (2015). PID control in C on an Arduino, with its electronics.
- Animated video explaining the Legendre transform (2022).

Awards

- UBC Graduate Departmental Award, 2021 and 2022.
- NSERC Undergraduate Student Research Award, 2020.
- Ernest Fox Award for Excellence in Mathematics, Marianopolis College, 2017 (one graduating student per year).
- Robohacks 2017: Best CEGEP and Most Imaginative Robot. McGameJam 2017: Special Mention.
- Canadian Computing Competition, top 25%, 2016 and 2017.
- Fermat Math Contest, school champion, 2015. AMQ Math Competition, 7th of 161, 2014.

Talks

- *Qualification-free duality via reduction to the joint supporting subspace*, West Coast Optimization Meeting, UW Tacoma, 2026.
- *Denoising guarantees for optimized sampling*, 5th Biennial SIAM Pacific Northwest Meeting, UW Seattle, 2025.
- *Exact regularization of linear programs*, 5th Biennial SIAM Pacific Northwest Meeting, UW Seattle, 2025.
- *Model-adapted Fourier sampling for generative compressed sensing*, NeurIPS Workshop on Deep Learning and Inverse Problems, New Orleans, 2023 (Oral).
- *Generative recovery from subsampled Fourier measurements*, CMS Winter Meeting, Montreal, 2023; UBC MILD seminar, 2022.
- Group talks: terminal embeddings (2022-23); convex sets and faces, optimal transport and conic duality (2026).

Service

- Referee, Journal of Optimization Theory and Applications (2026).
- Workshop on Obsidian and Zettelkasten note-taking for research, UBC Institute of Applied Mathematics (talk 2022, workshop 2024).

Skills

- **Programming:** Python (PyTorch, NumPy, pandas, Flask), Julia (Flux.jl), TypeScript/JavaScript, C#, Java; Git, Linux, Slurm, LaTeX, Typst.

Interests

- Singing (choirs, an a cappella group), acting (the musical *First Date*), Jiu-Jitsu.